Date: 30/07/2026

Lab Practical #08:

Study Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

Practical Assignment #08:

1. Give details about usage of Wireshark tool.

- Wireshark is a free and open-source network protocol analyzer used to capture and analyze network traffic in real time. It helps network administrators, cybersecurity professionals, and students monitor communication between devices and troubleshoot network-related problems.

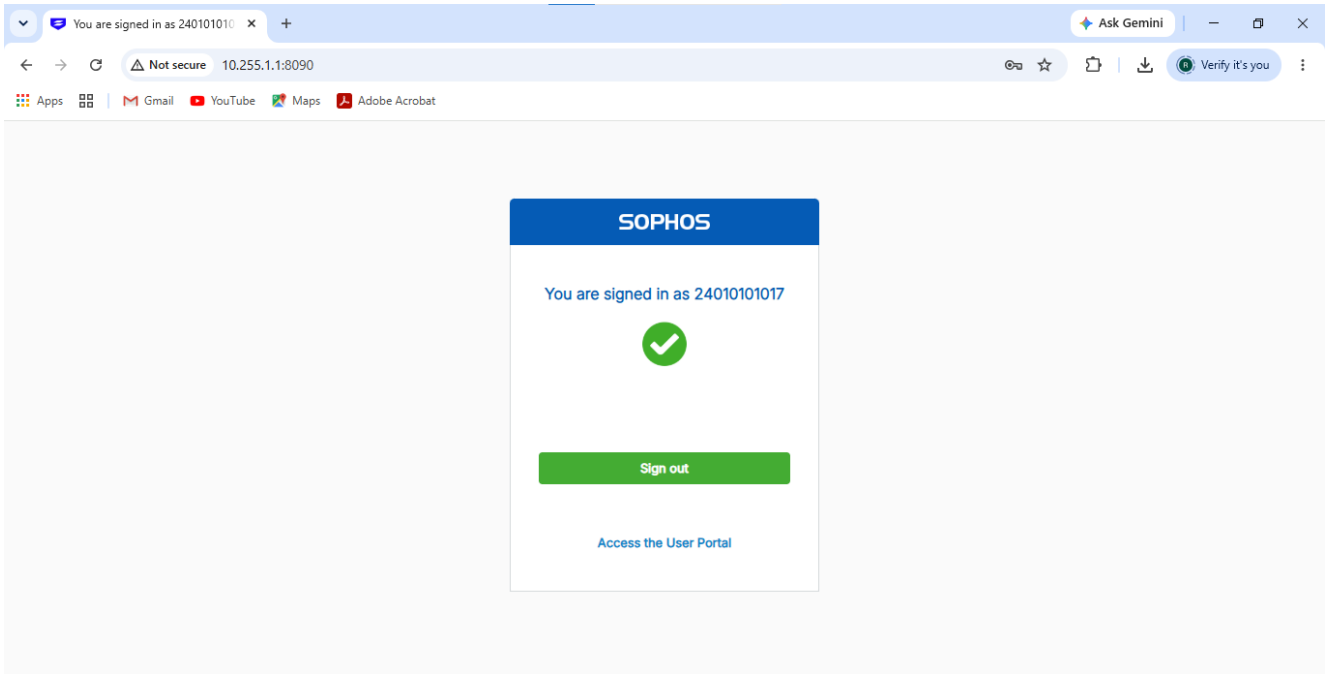
  Uses of Wireshark

  - Captures live network packets.
  - Analyzes different network protocols (HTTP, HTTPS, TCP, UDP, DNS, ICMP, etc.).
  - Troubleshoots network connectivity issues.
  - Detects suspicious or malicious network activity.
  - Monitors bandwidth and network performance.
  - Helps in cybersecurity investigations and packet analysis.
  - Used for learning networking protocols.

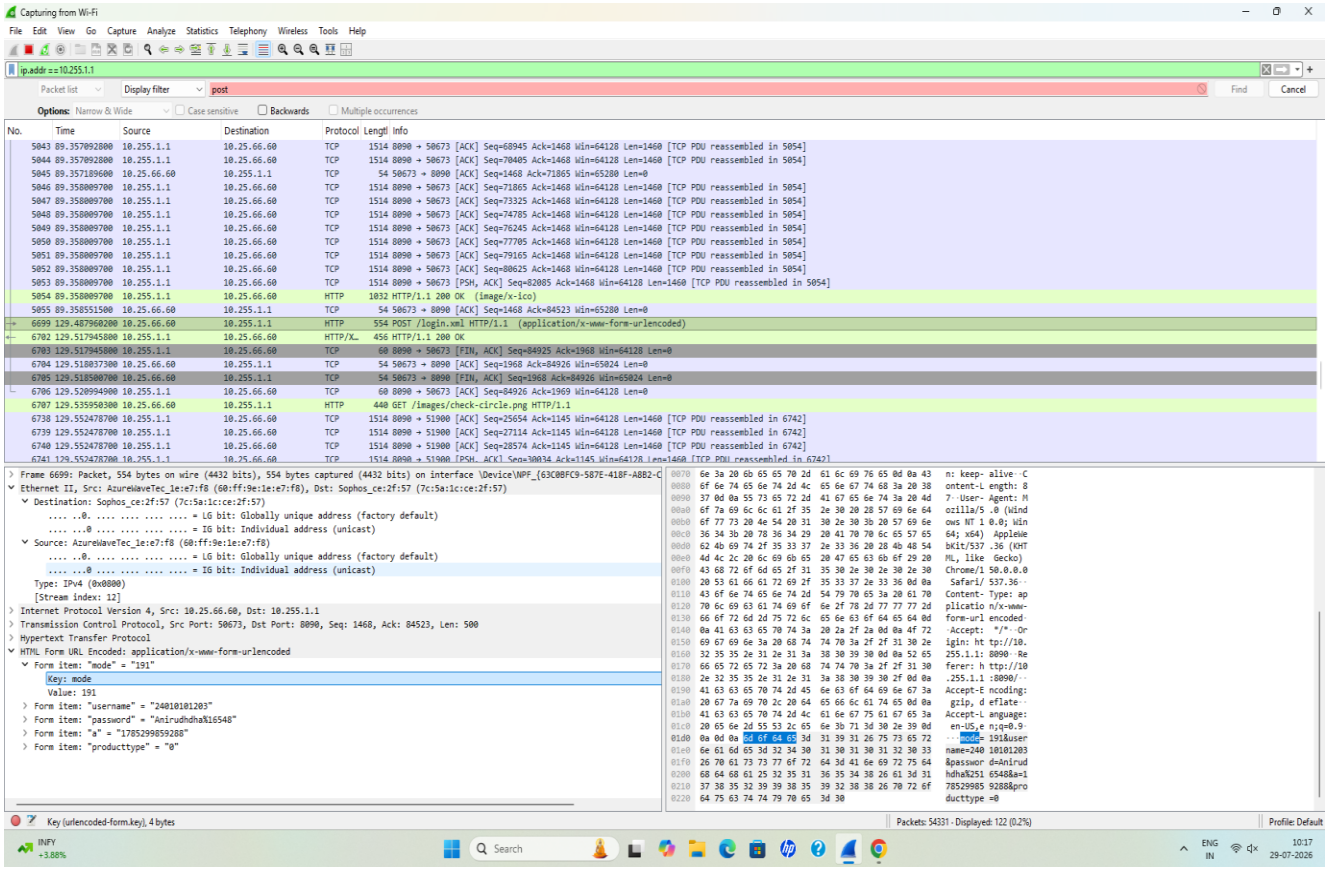
2. Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

1) Study Packet capture and header analysis by wire-shark.
   a) Capturing and observe Live Traffic – Use any website.
2) Applying Capture Filters:
   a) HTTP traffic: http
      i) For HTTP - Inspect request and response headers, URLs, and status codes.
      ii) For HTTPS, observe the handshake process (TLS).
   b) DNS requests: dns
      i) Observe how domain names are translated into IP addresses.
      ii) Identify DNS request and response packets.
   c) TCP packets: tcp
      i) Observe the 3-way handshake (SYN, SYN-ACK, ACK).
      ii) Identify sequence numbers and acknowledgment numbers.
   d) ICMP (ping): icmp
      i) Send ping requests (ping google.com) and observe request/reply packets.

Date: 30/07/2026

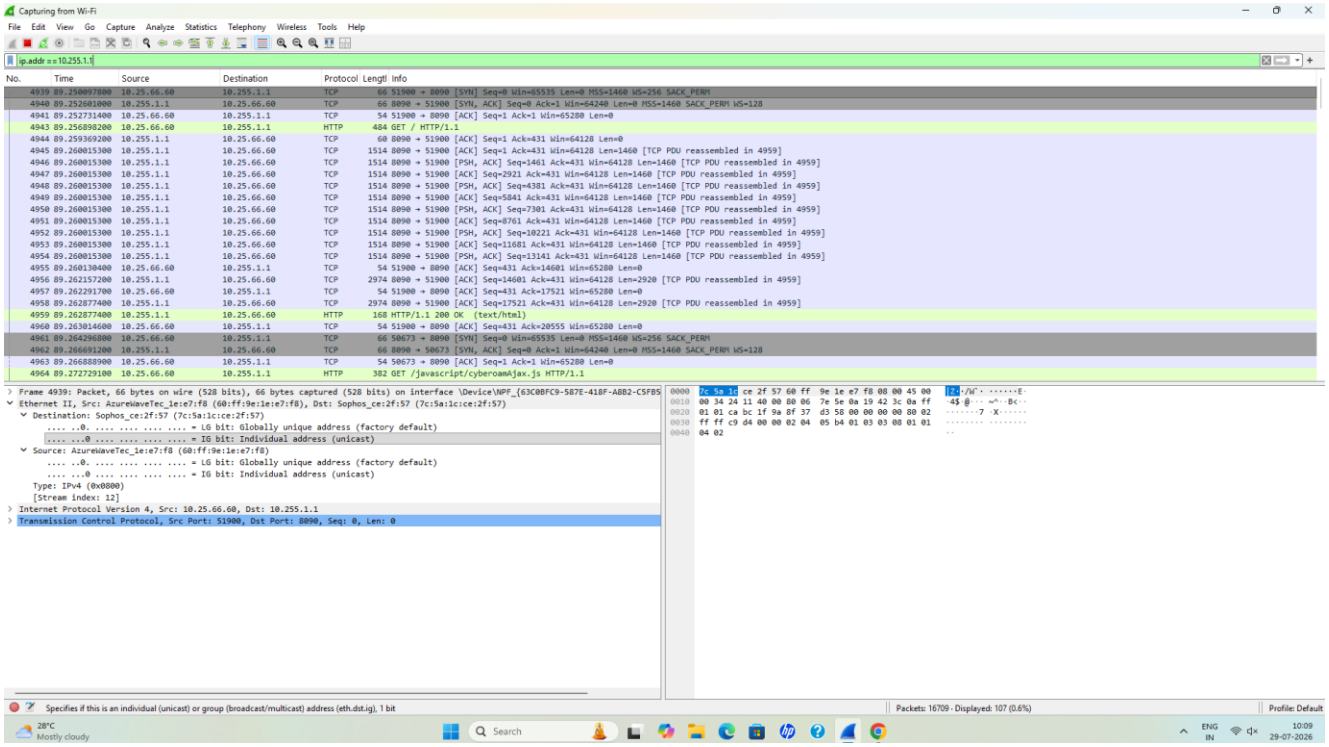


1) HTTP traffic: http ; Request And Response , Headers And URLs

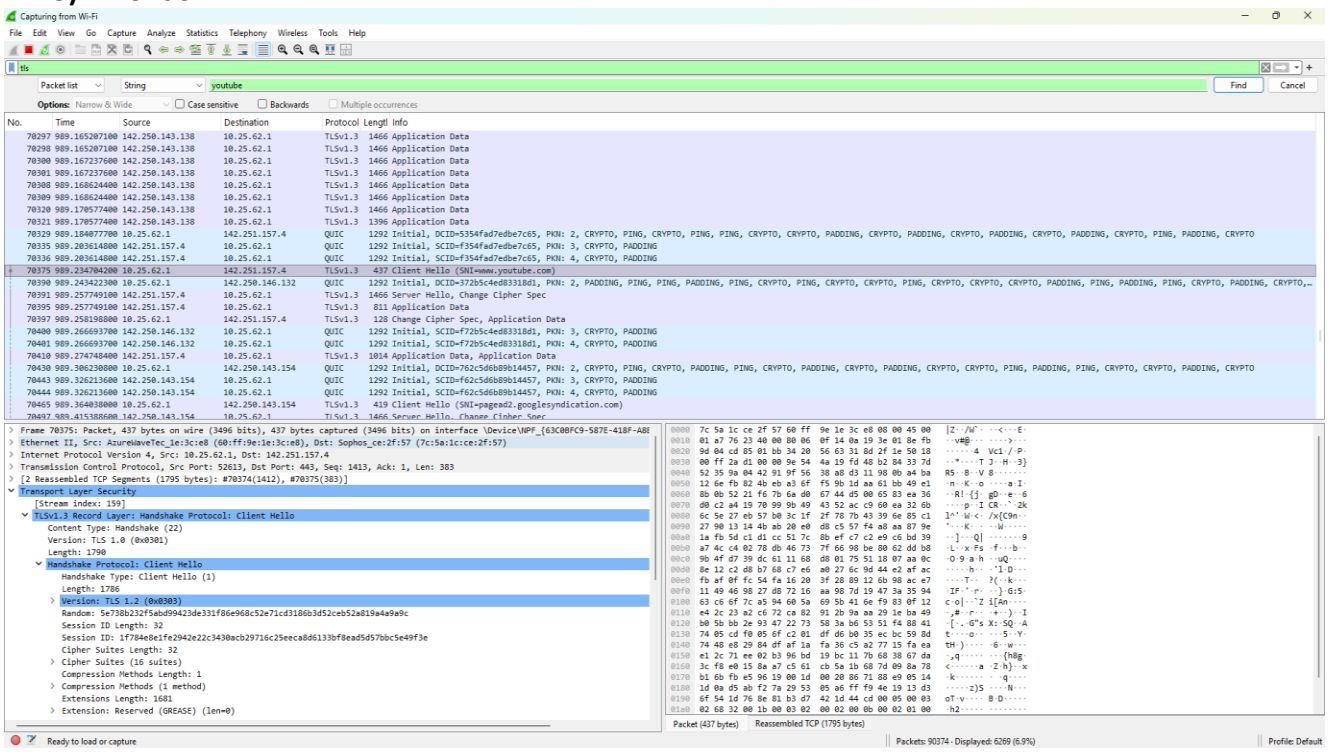


2) Capturing and observe Live Traffic – Use any website

Date: 30/07/2026

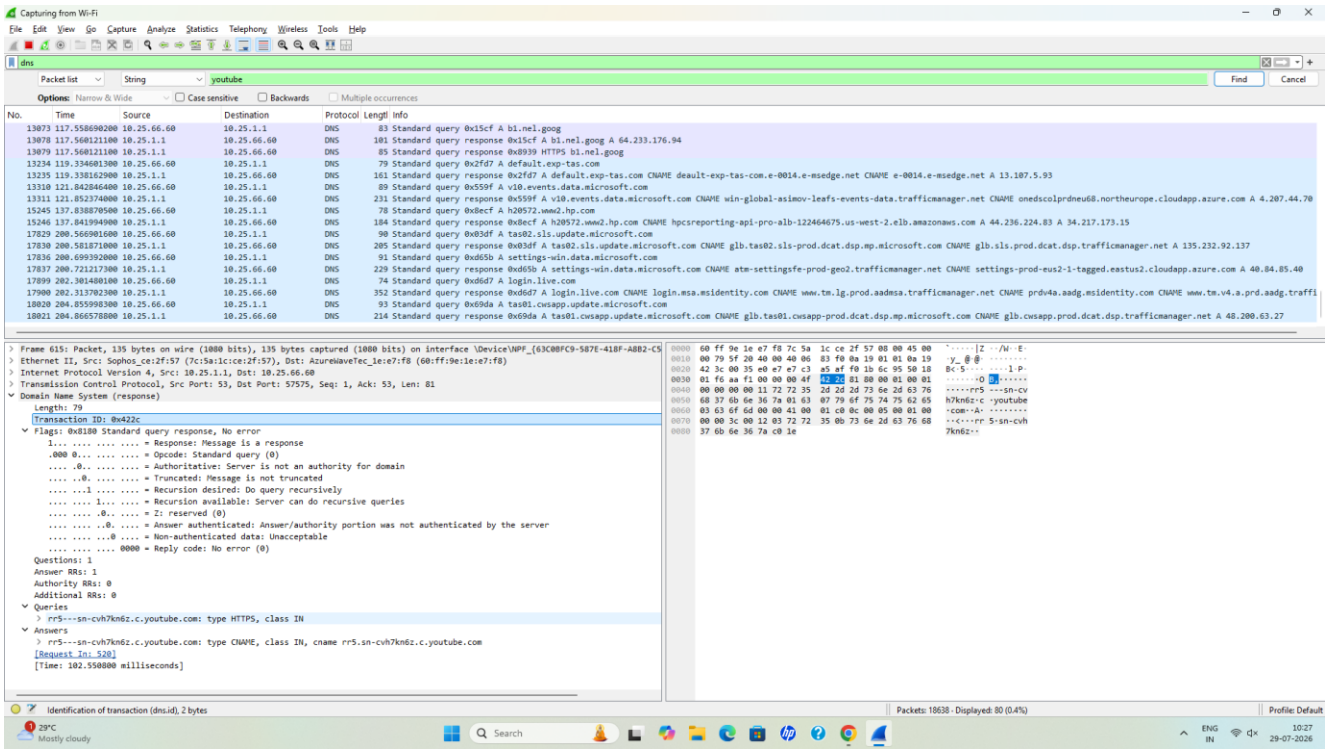


3) TLS: tls

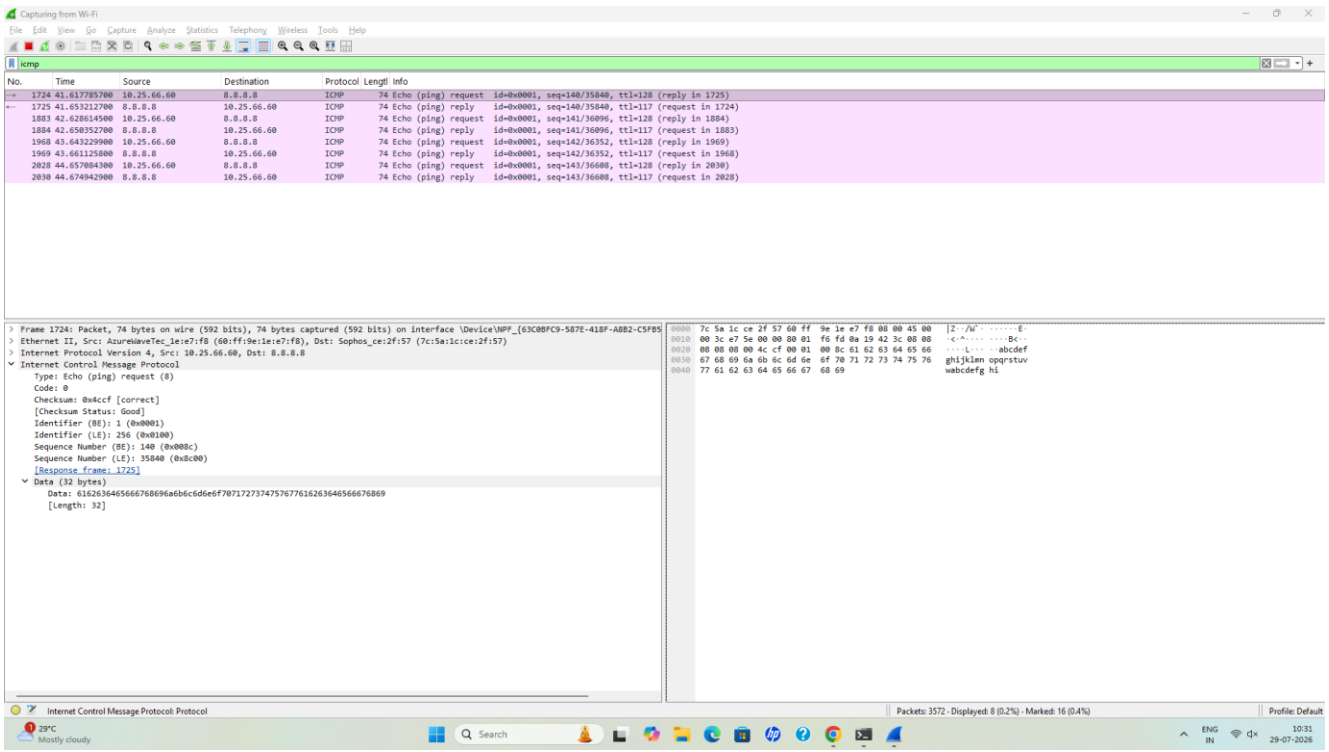


Date: 30/07/2026

1) DNS requests: dns



2) ICMP (ping): icmp



Date: 30/07/2026

3) TCP packets: tcp

